

6.1 Solving Linear Inequalities in One Variable

Lesson Introduction

 Benchmark	MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable, linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.		 Learning Targets	- I can solve two-step or multi-step one-variable inequalities. - I can graph the solution of an inequality on a number line. - I can interpret solutions to problems as viable or non-viable options.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.2.2 Solve two-step, linear inequalities in one variable, and represent solutions algebraically and graphically.		MA.912.AR.3.3 Given a mathematical or real-world context, write and solve one-variable, quadratic inequalities over the real number system. Represent solutions algebraically or graphically. MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable, absolute value inequalities. Represent solutions algebraically or graphically.	
 Essential Questions	- How do you solve a one-variable inequality? - How do you graph the solution of an inequality on a number line?			
 Lesson Narrative	In this lesson, students extend what they learned about solving one- and two-step inequalities in prior courses to solving multi-step inequalities. Students begin with a refresher on solving and graphing linear inequalities in one variable before moving into solving multi-step linear inequalities. The lesson extends to include linear inequalities representing real-world contexts. Within real-world contexts, students will determine whether solutions are viable or non-viable.			
 Component Summary	6.1.1 Warm-Up (~5 min)	Career Exploration: Planetary Research Scientist (<i>SE p. 291</i>)	6.1.3 Guided Instruction (15-20 min)	Solve multi-step, one-variable inequalities and relate inequalities to real-world contexts (<i>SE p. 293</i>)
	6.1.2 Refresher (5-10 min)	Refresher activity on solving and graphing linear inequalities in one variable (<i>SE p. 292</i>)	6.1.4 Collaboration (10-15 min)	Error Analysis and solve multi-step inequalities in one variable (<i>SE p. 295</i>)
	6.1.5 Wrap-Up (~5 min)	Growth Mindset reflection activity (<i>SE p. 297</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Warm-Up 6.1.1 contains a video to accompany the question prompts.	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to reverse the inequality symbol when multiplying or dividing by a negative value. - Students may confuse when to use an open or closed circle when graphing solutions on a number line. - Students may incorrectly translate the inequality from the word problem when phrases such as “no more than” and “at least” are used.	- Students have worked with inequalities in prior grade levels. Avoid adjusting problem values in the lesson to make the questions “easier,” as this could turn it into below grade level work. - Determine if your students need the support of all or part of the 6.1.2 Refresher. - If time is an issue, consider assigning part of the 6.1.4 Collaboration as homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson includes word problems in real-world contexts. Consider leveraging the Three Reads instructional routine to help students understand the word problem and what it is asking for, prior to beginning to solve it.	MA.K12.MTR.7.1: Real-World Contexts Activities 6.1.3 Guided Instruction and 6.1.4 Collaboration provide students with multiple problems involving the real-world application of linear inequalities. During these activities, students will connect this mathematical concept to everyday experiences and see how it can be applied to other disciplines.
 Differentiate Instruction	Consider converting the knowledge from 6.1.2 Refresher into an anchor chart that students can use as a reference when solving inequalities. Having a reference chart with the different scenarios can provide a “quick glance” support for students as they work through these problems. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

6.1.1 Warm-Up: Planetary Research Scientist

Component Overview: Students will explore the career of a planetary research scientist, Margarita Marinova, and then reflect on statements she made in the video.



6.1.1 Warm-Up: Planetary Research Scientist

Margarita Marinova is a planetary research scientist who studies Earth, as well as other planets. Margarita says the biggest key to success is to be passionate about something. She shares that while in high school, she got laughed at because she wanted to study Mars. Instead of reacting negatively to her peers, she founded the Toronto Chapter of the International Mars Society. Throughout her career, her passion for Mars has led her to spearheading many projects that focus on understanding Mars.

1. Why do you think it is important for people to be encouraged, rather than discouraged, about their passions?

Answers will vary.

Margarita says she loves asking the question “why?” She said one of the things that she thinks is important is being curious.

2. Describe something you have a passion and curiosity about that you think could lead you to a certain career.

Answers will vary.



For this activity, the students will watch a video of the career prior to reading the narrative and answering the questions. Consider having the video loaded in the tab of a browser prior to the start of class to avoid any potential technical issues and maintain control of the pacing of the lesson.

6.1.2 Refresher: Solving & Graphing Linear Inequalities in One Variable

Component Overview: Students will complete a refresher on how to solve and graph linear inequalities in one variable. Students will need a partner for this activity.



6.1.2 Refresher: Solving & Graphing Linear Inequalities in One Variable

1. Two inequalities are given in the table.

- a. Solve each inequality and then graph each solution set on the number lines provided.

Inequality A	Inequality B
$\frac{1}{2}(18k - 5) \geq 20$ $k \geq \frac{5}{2}$	$-\frac{1}{2}(18k - 5) \geq 20$ $k \leq \frac{-35}{18}$

- b. Check your work with your partner and if necessary, resolve any differences.

Answers will vary.

- c. Compare the given inequalities, the steps in solving each inequality, and the two solutions.

Similarities	Differences
<i>Answers will vary. Both inequalities require the use of the distributive property. Both require that you get rid of the $\frac{1}{2}$ first. Both inequalities are ≥ 20.</i>	<i>Answers will vary. For inequality B, the inequality sign is flipped in the process. Inequality B ends up with a $\leq$ and inequality A ends with a $\geq$.</i>

- d. Discuss with your partner, without solving, how the graphs of each solution can be found. Then, graph the solution set on each number line.

Inequality C	Inequality D
$\frac{1}{2}(18k - 5) > 20$	$-\frac{1}{2}(18k - 5) > 20$



The goal of this activity is to have students recall some key features about the process of solving and graphing one-variable inequalities:

- When and why to reverse the inequality symbol
- When to represent a solution on the number line as with an open circle vs a closed circle and what this means
- The direction of the arrow on the graph when representing the solution set on a number line



How can you use the values on the number line to check if your solution set is correct?

6.1.3 Guided Instruction: Multi-Step Linear Inequalities

Component Overview: Students will be introduced to solving multi-step, linear inequalities in one variable, and relating such inequalities and their solutions to real-world contexts.



6.1.3 Guided Instruction: Multi-Step Linear Inequalities

1. Mariana and Zion were solving the inequality shown.

$$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$$

Mariana and Zion used different first steps to solve the inequality. Their first steps are shown in the table.

Mariana	Zion
$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$	$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$
$\frac{3}{4}h - 2 \geq -\frac{20}{8}h + 24$	$-\frac{3}{16}h + \frac{1}{2} \leq \frac{5}{8}h - 6$
$\frac{26}{8}h \geq 26$	$6\frac{1}{2} \leq \frac{13}{16}h$
$h \geq 8$	$\frac{13}{2} \leq \frac{13}{16}h$
	$8 \leq h$

- Explain what Mariana did in her first step.
Mariana applied the distributive property to rewrite the expression.
- Explain what Zion did in his first step.
Zion divided both sides of the inequality by -4 and flipped the inequality symbol.
- Discuss with your partner whether Mariana and Zion will get the same answer based on their first steps. Summarize your discussion.
Answers will vary. Yes, Mariana and Zion should get the same answer because both of their first steps were valid.



In question 1, the goal is for students to see the flexibility in how to solve inequalities, and to solve one multi-step problem together as a class.

Each student should have all the strategies that they need from solving equations in Unit 5, and their prior knowledge of inequalities. Consider referencing the re-activated knowledge from the refresher activity as the questions are being discussed for each step.



For question 1b:

- Notice that the inequality symbol changed. Why is that?
- Why didn't it change for Mariana?

6.1.3 Guided Instruction: Multi-Step Linear Inequalities (continued)

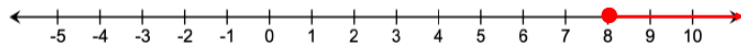
- d. Complete the solutions for both Mariana and Zion, continuing from their first step in the space provided in the table.

Answers in the table above.

- e. Write the solution set of the inequality in set-builder notation.

$\{h \in \mathbb{R} : h \geq 8\}$

- f. Graph the solution set of the inequality.



2. Use the inequality $0.04x + 1.02 > -0.27x + 17.14$ to complete the following.

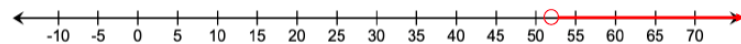
- a. Solve the inequality.

$$0.04x + 1.02 > -0.27x + 17.14$$

$$0.31x > 16.12$$

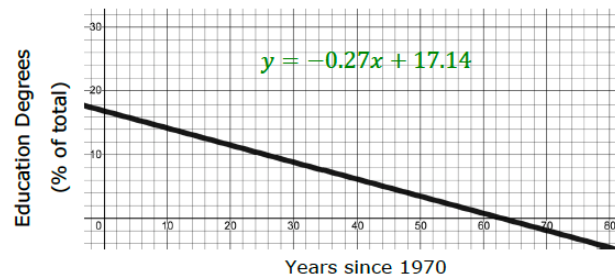
$$x > 52$$

- b. Graph the solution set.



3. The National Center for Education Statistics has gathered data on the majors of graduating college students since 1970. The percentage of degrees in education and computer science are shown.

Education Degrees Conferred (1970-2017)



Which method do you find to be the easiest for you to work with? Why?



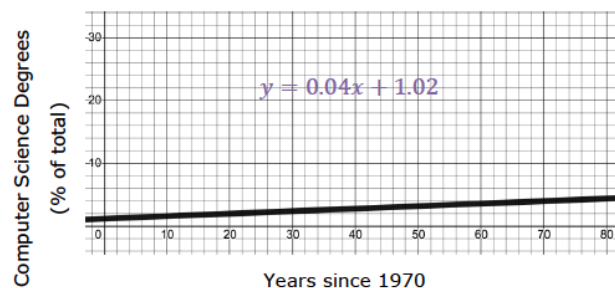
Note: There is not a correct answer or strategy to use. Different strategies may lend themselves better to particular inequalities based on their structure.



When beginning question 3, consider leveraging the Three Reads instructional routine by modeling it and having students follow along to reinforce the process. They could then apply the same strategy for the real-world context in the next component.

6.1.3 Guided Instruction: Multi-Step Linear Inequalities (continued)

Computer Science Degrees Conferred (1970-2017)



- a. Discuss with your partner how this context could be related to the inequality solved in the previous question, $0.04x + 1.02 > -0.27x + 17.14$.

Answers will vary.

The inequality $0.04x + 1.02 > -0.27x + 17.14$ can be used to compare the degrees conferred in education and degrees in computer science over time. The table gives information about the meaning of the mathematical expressions contained in the inequality in this context.

Mathematical Expression	Real-World Context
x	The number of years that have passed since 1970
$0.04x + 1.02$	The expected percentage of total degrees conferred that are computer science degrees
$-0.27x + 17.14$	The expected percentage of total degrees conferred that are education degrees



In the first year that this data was collected, education was the most popular major, with approximately 21% of students graduating with a degree in education. On the other hand, computer science was one of the least popular majors, with only approximately 0.3% of degrees conferred.

In recent years, the percentage of students majoring in education has been steadily decreasing, while the percentage of students majoring in computer science is on the rise.



Question 1a: This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine what students understand about this context.

The two expressions on each side of the inequality sign represent the percentage of computer science degrees and percentage of education degrees. This inequality represents the comparison of the years in which the percentage of computer science degrees conferred is greater than that percentage of education degrees conferred.

6.1.3 Guided Instruction: Multi-Step Linear Inequalities (continued)

- b. Which of the following questions could be answered by using the solution set of the inequality $0.04x + 1.02 > -0.27x + 17.14$?

- For which number of years after 1970 was the percentage of total degrees in computer science greater than that in education?
- For which number of years after 1970 was the percentage of total degrees in education greater than that in computer science?

- c. Refer back to your solution to this inequality. What does the solution set tell you about the degrees conferred in education compared to degrees conferred in computer science?

Answers will vary. The solution set indicates that for any year after 2022 there will be more Computer Science degrees than Education degrees.

- d. Discuss with your partner whether 79.3 is a viable solution in this real-world context. Summarize your discussion.

Answers will vary. Yes, it is a viable solution because it is greater than 52 and the decimal represents a partial year, which is also viable.



What would be an inequality that could be used to represent the other answer choice in part b?



At the conclusion of this activity, engage students in a whole-group discussion to highlight the following:

- *Non-integral solutions are viable because it is possible to have part of a year.*
- *Any solution to the mathematical inequality is viable in this context since they are all positive.*

6.1.4 Collaboration: Multi-Step Inequalities in One Variable

Component Overview: Students will work with a partner to solve multi-step inequalities in one variable in both mathematical and real-world contexts.



6.1.4 Collaboration: Multi-Step Inequalities in One Variable

1. Marquis made an error when solving the inequality $\frac{-82b+10}{5} - 35 < 90$. His work is shown.

Given Inequality	$\frac{-82b+10}{5} - 35 < 90$
Step 1:	$-82b + 10 - 35 < 450$
Step 2:	$-82b - 25 < 450$
Step 3:	$-82b < 475$
Step 4:	$b > -\frac{475}{82}$

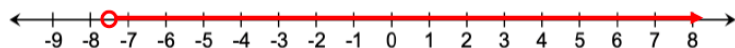
- a. With your partner, identify the step in which Marquis made an error. Write a note to Marquis explaining why that step was incorrect.
Step 1 was incorrect because when multiplying both sides by 5, Marquis only multiplied the first term on the left side by 5 (i.e., he forgot to multiply -35 by 5) but he should have multiplied both terms by 5.

- b. Solve the inequality $\frac{-82b+10}{5} - 35 < 90$.

$$\begin{aligned} \frac{-82b+10}{5} &< 125 \\ -82b+10 &< 625 \\ -82b &< 615 \\ b &> -7.5 \end{aligned}$$

- c. Write the solution set in set-builder notation.
 $\{b \in R : b > -7.5\}$

- d. Graph the solution set.



For this activity, students should work with a partner or small group to attempt both problems from start to finish without modeling or support.



MA.K12.MTR.7.1: Discussion and Reflection
 Question 1 provides an opportunity for students to engage with MA.K12.MTR.7.1. Students are analyzing the mathematical thinking of others, recognizing errors, and suggesting how to correctly solve the problem.

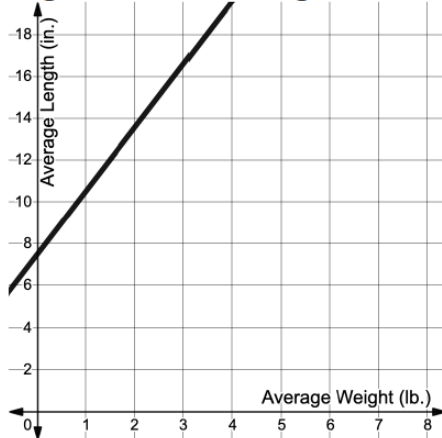
6.1.4 Collaboration: Multi-Step Inequalities in One Variable (continued)

2. Lake Talquin in northern Florida is known to have Calico Bass, a freshwater species that prefers warm climates. The equation $l = 3.08w + 7.844$ models the relationship between the length, l , and the weight, w , of the average Calico Bass. The graph of the relationship is shown.



For struggling readers, consider facilitating the Three Reads process with prompting questions for each read based on an identified purpose for each read. This can help their comprehension of the problems and meta-awareness of mathematical language.

Average Calico Bass Length vs. Weight



Any Calico Bass caught in Lake Talquin that is less than 10 inches (in.) long must be released immediately. The inequality $3.08w + 7.844 \geq 10$ can be used to determine whether a Calico Bass can be kept once caught.

- a. Solve the inequality $3.08w + 7.844 \geq 10$.
 $3.08w \geq 2.156$
 $w \geq 0.7$

6.1.4 Collaboration: Multi-Step Inequalities in One Variable (continued)

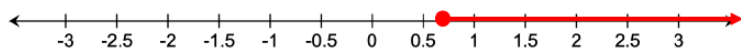
- b. Explain what the solution set represents in the context.

Answers will vary. The solution set means that any Calico Bass that weighs at least 0.7 pounds can be kept and doesn't need to be released back into the water.



When circulating, listen for student responses to question 2b. Students may likely reverse the context or simply add the words without putting together a coherent sentence that articulates what it represents in context – be aware of this misconception and address when necessary.

- c. Graph the solution set.



Consider asking the following probing questions as needed:

- Based on what you wrote for question 2b, does 0.58 pounds meet the criteria for her to keep the fish?
- Have you been fishing where you had limits on the size of fish you can keep?

- d. Lucia was fishing in Lake Talquin and caught a Calico Bass that weighed 0.58 pounds. Use the solution set to explain whether or not Lucia needs to release the fish.

Answers will vary. Because 0.58 is not in the solution set, Lucia needs to release the Calico Bass back into the water.

- e. Discuss with your partner whether there are any solutions to the mathematical inequality that are not viable in the real-world context. Summarize your discussion.

Answers will vary. The mathematical solution set includes all values greater than or equal to 0.7 pound. The weight of a Calico Bass fish has an upper limit that isn't given in the problem but exists in the real world. For example, it would not be realistic for a Calico Bass to weigh 30 pounds.

6.1.5 Wrap-Up: Growth Mindset

Component Overview: Students will engage in a Growth Mindset activity about what to do when you feel like quitting by first reading a quote from Thomas Edison.



6.1.5 Wrap-Up: Growth Mindset

"Many of life's failures are people who did not realize how close they were to success when they gave up." - Thomas Edison

1. If you start to feel like quitting, what will you do in that moment to persevere?

Answers will vary.

2. What would you want your teacher to say to you when they see you taking a risk, trying hard, or pushing through mistakes to pursue this goal?

Answers will vary.



Consider using what students write for question 2 in the next lesson.